

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** House Rent
- **Team Members:** N. Chaithanya Krishna – Project Development, Documentation, Testing & Presentation

2. Project Overview

- **Purpose:** The House Rent project is a web-based rental management system designed to simplify the process of renting residential properties. It provides a centralized platform where renters can search and book verified properties, owners can manage property listings and bookings, and administrators can oversee the entire system efficiently.
- **Features:**
 - User Registration & Login
 - Role-Based Authentication (Admin, Owner, Renter)
 - Property Upload & Management
 - Property Search & Filters
 - Property Booking
 - Booking Approval / Rejection
 - Admin Dashboard
 - Responsive UI
 - Dark/Light Mode

3. Architecture

- **Frontend:** The frontend is developed using React.js, HTML5, CSS3, JavaScript, and Bootstrap to create a responsive and user-friendly interface.
- **Backend:** The backend is developed using Node.js and Express.js, providing RESTful APIs, authentication, booking management, and business logic.
- **Database:** MongoDB stores user information, property details, booking records, and transaction history.

4. Setup Instructions

- **Prerequisites:**
 - Node.js
 - npm
 - MongoDB
 - Visual Studio Code

- **Installation:**
git clone <repository-url>
cd frontend
npm install
cd ../backend
npm install
Create the required `.env` file and configure the MongoDB connection.

5. Folder Structure

- **Frontend:**

frontend/

├── src/

├── modules/

├── components/

├── App.js

└── package.json

- **Backend:**

backend/

├── routes/

├── models/

├── controllers/

├── middleware/

├── index.js

└── package.json

6. Running the Application

- Provide commands to start the frontend and backend servers locally.

- **Frontend:**

cd frontend

npm start

- **Backend:**

cd backend

npm start

7. API Documentation

Method	Endpoint	Description
POST	/register	Register User
POST	/login	Login User
GET	/properties	View Properties
POST	/property	Add Property

Method	Endpoint	Description
PUT	/property/:id	Update Property
DELETE	/property/:id	Delete Property
POST	/booking	Book Property
GET	/bookings	View Bookings

8. Authentication

- Authentication is implemented using **JWT (JSON Web Token)**. Passwords are encrypted using **bcrypt.js**, and role-based authorization is provided for Admin, Owner, and Renter.

9. User Interface

Home Page:



Register or Sign Up:

HomeSphere

 [Home](#) [Login](#) [Register](#)



Sign Up

User Type

SIGN UP

Already have an account? [Sign In](#)

Login:

HomeSphere

 [Home](#) [Login](#) [Register](#)



Sign In

SIGN IN

[Forgot Password? Click here](#)

[Don't have an account? Sign Up](#)

Renter Dashboard:

HomeSphere

Hi Ramesh Log Out

All Properties

Booking History

Transactions

Filter By: :Address

All Ad Types

All Types



Location:
Banglore

Property Type:
residential

Ad Type:
rent

Owner Contact:
9871234056

Availability:
Available

Property Amount:
Rs.2000000

Get More Info of the Property

Get Info



Location:
Tirupati

Property Type:
residential

Ad Type:
sale

Owner Contact:
9874560123

Availability:
Available

Property Amount:
Rs.3000000

Get More Info of the Property

Get Info



Location:
Tirupati

Property Type:
land/plot

Ad Type:
sale

Owner Contact:
9874561203

Availability:
Available

Property Amount:
Rs.4000000

Get More Info of the Property

Get Info

Booking Page:

Transactions

Filter By: :Address

All Ad Types

All Types

Property Info



Property Details

Owner Contact : 9871234056

Location Details

Address : Banglore

Transactions

Filter By: /Address /All Ad Types /All Types

Property Info

Property Details

Owner Contact : 9871234056

Availability : Available

Property Amount : ₹2000000

Location Details

Address : Bangalore

Property Type : residential

Ad Type : rent

Additional Information

Additional Info:

Confirm Your Booking

Full Name

Full Name

Phone Number

0

Book Property

© 2020 Copyright, HomeSphere

Booking History:

HomeSphere

Hi Ramesh Log Out

All Properties

Booking History

Transactions

Booking ID	Property ID	Tenant Name	Phone	Booking Status	Actions
6a4a9987fea2c59308fcdeb8	6a4a9601fea2c59308fcde98	Ramesh	9987654321	pending	

Admin Login:

Admin Login

Sign in with your admin credentials

Admin Email

Password

Login

Admin Dashboard:

HomeSphere							Hi Admin
User ID	Name	Email	Type	Granted (Old Logic)	isApproved (New)	Actions	
6a4348a781b44fb517f920c3	Admin	chaithuyadav03@gmail.com	Admin	granted	☒	<button>TOGGLE GRANTED</button>	<button>UNAPPROVE</button> <button>DELETE</button>
6a4a9517fea2c59308fcde7f	Ramesh	Ramesh@gmail.com	Renter	ungranted	☒	<button>TOGGLE GRANTED</button>	<button>UNAPPROVE</button> <button>DELETE</button>
6a4a954bfea2c59308fcde8b	Mahesh	Mahesh@gmail.com	Owner	ungranted	☒	<button>TOGGLE GRANTED</button>	<button>UNAPPROVE</button> <button>DELETE</button>

Owner Dashboard:

HomeSphere			Hi Mahesh	Log Out
Add Property All Properties All Bookings Transactions	Property type Property Ad type Property Full Address Residential Rent Address Property Images Owner Contact No. Property Amt. Choose Files No file chosen contact number 0 Additional details for the Property Submit form			

Owner Bookings:

HomeSphere						Hi Mahesh	Log Out
Add Property All Properties All Bookings Transactions	Booking ID	Property ID	Tenant Name	Tenant Phone	Booking Status	Actions	
	6a4a9987fea2c59308fcdeb8	6a4a9601fea2c59308fcde98	Ramesh	9987654321	pending	<button>Change</button>	

10. Testing

- The application was tested using manual testing to verify user registration, login, property management, booking workflow, transaction history, authentication, and responsive design.

11. Screenshots or Demo

https://drive.google.com/file/d/1PTmUI0McrIAjWDb_kf_tLbWpo7Va6SFX/view?usp=sharing

12. Known Issues

- Currently, no major issues were identified during testing. Minor UI enhancements and additional features may be implemented in future versions

13. Future Enhancements

- Online Payment Gateway
- Google Maps Integration
- AI-Based Property Recommendation
- Chatbot Support
- Email & SMS Notifications
- Wishlist Feature
- Mobile Application
- Property Rating & Reviews